

***Generation Interconnection  
System Impact Study Report***

***For***

***PJM Generation Interconnection Request  
Queue Position AE1-001***

***Calvert Cliffs 500kV  
7.1 MW Capacity / 28.1 MW Energy***

Revised: October 2022

Revised: April 2021

Issued: February 2021

## Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, Section 205, as well as the System Impact Study Agreement between Calvert Cliffs Nuclear Power Plant LLC, the Interconnection Customer (IC) and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Baltimore Gas and Electric Company.

## Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the IC. As a requirement for interconnection, the IC may be responsible for the cost of constructing Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an IC may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The IC is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

## Revision History

The April 2021 report updates the project capability table included in the previous issue.

The October 2022 revision updates the Stability and Reactive Power Requirement section of this study.

## General

The Interconnection Customer, Calvert Cliffs Nuclear Power Plant, LLC, has proposed an uprate to the existing Calvert Cliffs nuclear facility located in Calvert County, Maryland. Queue AE1-001 project is a proposal to increase the output of Unit 1 by 28.1 MW (7.1 MW Capacity). The previous queue requests for this facility were M04, AC1-008 and AC2-017. The IC proposed in service date for this project of August 2019.

Calvert Cliffs Unit 2 will remain at 861.0 MW of Capacity Interconnection Rights. As a result of the AE1-001 request, the Capacity Interconnection Rights for Unit 1 will increase to **884.2 MW**. The Maximum Facility Output of Unit 1 as a result of this uprate is **921.5 MW**.

The project capability is summarized below:

|                    | Summer (Net Energy) |            |               |
|--------------------|---------------------|------------|---------------|
|                    | Unit 1              | Unit 2     | Unit 1 & 2    |
| Existing: M04      | 873                 | 875        | 1748          |
| AC1-008 Increase   | 0                   | 15         | 15            |
| Sub Total          | 873                 | 890        | 1763          |
| AC2-017 Increase   | 11.1                | 0          | 11.1          |
| Sub Total          | 884.1               | 890        | 1774.1        |
| AE1-001 Increase   | 28.1                | 0          | 28.1          |
| <b>Grand Total</b> | <b>912.2</b>        | <b>890</b> | <b>1802.2</b> |

|                    | Winter (Net Energy) |              |               |
|--------------------|---------------------|--------------|---------------|
|                    | Unit 1              | Unit 2       | Unit 1 & 2    |
| Existing: M04      | 900                 | 893          | 1793          |
| AC1-008 Increase   | 0                   | 6.3          | 6.3           |
| Sub Total          | 900                 | 899.3        | 1799.3        |
| AC2-017 Increase   | 11.1                | 0            | 11.1          |
| Sub Total          | 911.1               | 899.3        | 1810.4        |
| AE1-001 Increase   | 10.4                | 0            | 10.4          |
| <b>Grand Total</b> | <b>921.5</b>        | <b>899.3</b> | <b>1820.8</b> |

|                    | Capacity Interconnection Rights |            |               |
|--------------------|---------------------------------|------------|---------------|
|                    | Unit 1                          | Unit 2     | Unit 1 & 2    |
| Existing: M04      | 866                             | 841.8      | 1707.8        |
| AC1-008 Increase   | 0                               | 19.2       | 19.2          |
| Sub Total          | 866                             | 861        | 1727          |
| AC2-017 Increase   | 11.1                            | 0          | 11.1          |
| Sub Total          | 877.1                           | 861        | 1738.1        |
| AE1-001 Increase   | 7.1                             | 0          | 7.1           |
| <b>Grand Total</b> | <b>884.2</b>                    | <b>861</b> | <b>1745.2</b> |

### **Point of Interconnection**

AE1-001 will interconnect with the ITO transmission system via the Calvert Cliffs 500 kV substation. This is an existing facility and the Point of Interconnection is unchanged.

### **Cost Summary**

The AC2-017 project will be responsible for the following costs:

| Description                                           | Total Cost |
|-------------------------------------------------------|------------|
| Attachment Facilities                                 | \$0        |
| Direct Connection Network Upgrades                    | \$0        |
| Non Direct Connection Network Upgrades                | \$0        |
| New System Upgrades                                   | \$0        |
| Contribution to previously identified system upgrades | \$0        |
| <b>Total Costs</b>                                    | <b>\$0</b> |

## **Transmission Owner Scope of Work**

There is no transmission owner scope to accommodate this request.

### **Network Impacts**

The Queue Project AE1-001 was evaluated as a 28.1 MW (Capacity 7.1 MW) uprate to the AC2-017 project at the Calvert Cliffs substation in the BGE area. Project AE1-001 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-001 was studied with a commercial probability of 100%. Potential network impacts were as follows:

## **Summer Peak Analysis - 2022**

### **Generator Deliverability**

*(Single or N-1 contingencies for the Capacity portion only of the interconnection)*

None

### **Multiple Facility Contingency**

*(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)*

None

### **Contribution to Previously Identified Overloads**

*(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)*

None

### **Steady-State Voltage Requirements**

*(Results of the steady-state voltage studies should be inserted here)*

None

### **Short Circuit**

*(Summary of impacted circuit breakers)*

None

## **Affected System Analysis & Mitigation**

None.

### **Delivery of Energy Portion of Interconnection Request**

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request. Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

### **Summer Peak Load Flow Analysis Reinforcements**

#### **New System Reinforcements**

*(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)*

None

#### **Contribution to Previously Identified System Reinforcements**

*(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)*

*(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)*

None

### **Light Load Analysis – 2022**

No impacts identified.

### **Light Load Load Flow Analysis Reinforcements**

#### **New System Reinforcements**

*(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)*

None

#### **Contribution to Previously Identified System Reinforcements**

*(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)*

*(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)*

None

### **Stability and Reactive Power Requirement**

Generator Interconnection Request AE1-001 is for an increase in energy injection capability of the AC2-017 project at the Calvert Cliff station. The uprate increases the Maximum Facility Output (MFO) of the plant from 911.1 MW to 921.5 MW for Calvert Cliff Unit 1. AE1-001 has a Point of Interconnection (POI) at the existing Calvert Cliffs 500 kV Substation in the BGE transmission system, Calvert County, Maryland.

The power flow scenario for the analysis was based on the RTEP 2020 light load case, modified to include applicable queue projects. AE1-001 has been dispatched online at maximum facility output, with approximately unity power factor at the high side of the station transformer.

AE1-001 will be tested for compliance with NERC, PJM, and other applicable criteria. The study will include 95 contingencies, each with a 20 second simulation time period. Contingencies to be studied include:

- Steady-state operation (20 second simulation)
- Three-phase faults with normal clearing time
- Single-phase faults with a stuck breaker with delayed clearing time
- Single-phase faults with loss of multiple circuits caused by a common tower contingency

Other applicable criteria tested include:

- Transmission Owner (TO) specific criteria
- Nuclear Power Interface Requirement (NPIR) and Other criteria

The 95 fault contingencies tested on the 2020 light load case met the recovery criteria:

- The AE1-001 generators were able to ride through the faults except for faults where protective actions trip one or more generator(s).
- All generators maintained synchronism and any post-contingency oscillations are positively damped with a damping margin of at least 3%.

- All bus voltages recover to 0.7 p.u. within 2.5 seconds and the final voltages are within the steady-state voltage ranges below per Exelon-planning criteria.
  - P1, P2, P4, P5, and P7 Category Contingencies:
    - 0.92 to 1.05 p.u. for 230 kV or below
    - 0.97 to 1.10 p.u. for 500 kV
- No transmission element trips, other than those either directly connected or designated to trip as a consequence of the fault

The AE1-001 queue project did not meet the 0.9 lagging power factor requirement. An additional 34.12 MVAR and 12.86 MVAR reactive power is required for the project to meet the 0.9 lagging power factor requirement and at the generator's terminals for winter and summer gross outputs, respectively.

## System Configuration

